Juan Camacho Mohedano

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Research Interests

VLMs, LLMs, LLM agents: evaluation, interpretability and training

Education

Sep 2023 – Jul 2026 **University of Trento**, *MSc in AI Systems.*
Thesis: *What Do Large Multimodal Models Really Know? Unlocking Knowledge Through Test-Time Computation in Open-World Image Classification.*
Advisor: *Elisa Ricci.*

- Built a 256-sample evaluation framework on the OVEN benchmark separating coverage from reliability; showed a lenient LLM judge inverts the ranking of Qwen3-VL 2B–32B, while a specificity-preserving audit restores it.
- Extended the rsbench benchmark suite to Raven's Progressive Matrices with a DeepProbLog solver, exposing a reasoning shortcut: 0.91 answer F1 while two of three concepts stayed at chance, removed with 1% concept supervision.
- Compared GPT-2 expert neurons against fMRI recordings from nine subjects using representational similarity analysis; expert neurons align with brain activity better than the dense embeddings containing them (mean $d = 0.63$).

2019 – 2022 **University of Leeds**, *BSc Computer Science with AI.*
First class. Co-led the robotics team competing at RoboCup.
Thesis: *6DoF Robot Grasping.* Advisors: *Matteo Leonetti & Mehmet Dogar.*

2017 – 2019 **Marshalltown Community College**, *Associate of Arts.*

Work Experience

May 2025 – Jul 2025 **COVAP**, Pozoblanco, Spain.
AI Engineer
Built a CrewAI agentic workflow that cut budget-document validation from 30 minutes to under 5, and a multimodal chain validating export documentation against product-tag photographs. Shipped two OpenWebUI filters and evaluated GraphRAG, docling, OCRFlux and NuExtract for the document-ingestion pipeline.

2024 – 2025 **Freelance**, *ML / AI Engineer*
Benchmarked 13 log parsers and grew PyOD anomaly detection from 5 to 49 algorithms.
Customized RAGApp vector search and authored five applied AI notebooks.

Selected Projects & Open-Source

2026 **Wildfire Response Environment.** Built a decision-making environment for wildfire response.

2026 **OpenEnv.** Contributing managed, non-blocking child queries for Recursive Language Models.

Jan 2023 – Jun 2023 **NEURA Robotics**, Bielefeld, Germany.
Robotics Developer
Implemented ROS 2 servers for low-level control automation, explored state-of-the-art grasping techniques, and collected an RGB dataset to train a YOLO model for object detection.

Skills

Coding Python [PyTorch, Transformers, TRL, vLLM], SLURM, Git, Linux

Languages Spanish, English

Competitions & Awards

2026 **HackSpain.** *HappyRobot track winner in its first edition.*

2026 **CloudRift compute grant.** For RLM post-training.

2026 **EuroHPC cluster compute grant.** For my master's thesis.

2024 **NASA International Space Apps,** *honorable mention*; **Hub Innovazione Trentino Challenge; AI Scholarship, COVAP**

2020 -- 2022 **RoboCup 2021--22; RoboCup@Home Education,** *Gold Award* **2020; SciRoc 2021**